

CEN 300: Practical Problem

Project Report

Visual Attention Behavior of Distracted Pedestrians

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1 Introduction

Pedestrian safety remains a critical concern in road safety across the globe due to the rising number of fatalities resulting from road crashes. Among various traffic scenarios, road crossing is particularly complex as it involves dynamic and continuous interactions with the surrounding environment. A key aspect influencing the behavior of road users is their ability to anticipate the actions of others on the road. Hence, effective communication between road users plays a vital role in ensuring smooth and conflict-free mobility.

Road users utilize various communication techniques to interpret the intentions of others, broadly categorized into two types: explicit and implicit. Explicit communication involves deliberate actions that follow predefined traffic norms, such as using turn indicators, brake lights, honking, or lane change signals. In contrast, implicit communication consists of informal, non-verbal cues such as eye contact, vehicle speed changes, hand gestures, head movements, and facial expressions. Among these, eye contact or gazing behavior has been recognized as one of the most prominent implicit cues, especially in pedestrian-vehicle interactions. In fact, in conventional road-crossing scenarios, a significant proportion of pedestrians attempt to make eye contact with drivers to ensure safe passage.

Visual cues like gaze direction and duration are essential in helping pedestrians make safe crossing decisions. These cues are directly linked to situational awareness—a concept defined as the perception, comprehension, and projection of environmental elements over time. Situational awareness is critical for decision-making, and its absence often leads to human errors, which can result in accidents. In high-risk environments, such as roads, reduced situational awareness due to distraction can severely compromise safety.

One of the most influential factors affecting gaze behavior is the area of interest (AOI) in the pedestrian's field of view. Different age groups and individuals display varying gaze patterns depending on what they focus on—whether it's the road, traffic lights, vehicles, or signage. The presence of moving vehicles has been shown to significantly increase the duration of visual fixations, indicating heightened attention.

Distractions from multitasking activities such as mobile phone use or engaging with roadside billboards can significantly impact a pedestrian's gaze behavior and situational awareness. These distractions increase visual dispersion and reduce focus on traffic-related hazards. Activities like reading, texting, talking on the phone, or listening to music have all been linked to impaired hazard detection and delayed responses, as they draw attention away from the environment.

Fixation-related metrics provide meaningful data for evaluating attention and cognitive load. Metrics such as Total Fixation Duration (TFD), Fixation Count (FC), and Average Fixation Duration (AFD) are particularly useful. While TFD represents the cumulative attention given to an AOI, FC shows how often the AOI was revisited. AFD combines both TFD and FC to provide a clearer picture of the cognitive processing involved, making it a suitable metric for assessing situational awareness under complex scenarios like road crossing.



Although previous studies have mainly focused on distractions caused by mobile phone use, few have explored the impact of billboards, which are commonly located near intersections and are designed to capture the attention of road users. Considering both mobile devices and billboards as potential sources of distraction is essential for a comprehensive understanding of pedestrian behavior.

Most past research has used statistical tests to compare gaze behavior under different conditions. However, such tests are limited in their ability to quantify the combined effects of multiple variables. Therefore, this study aims to model Average Fixation Duration to better understand the influence of various factors on situational awareness during pedestrian road-crossing under the influence of multitasking activities.

2 Methodology

This study uses an advanced pedestrian simulator approach to assess the influence of different multi-tasking activities on the situational awareness of pedestrians while crossing the road without endangering the safety of pedestrians. The experimental setup and procedure details are described in the following sections.

2.1 Experimental Setup of Pedestrian Simulator

The experimental setup was designed to simulate real-world pedestrian crossing scenarios in a controlled virtual environment. The goal was to evaluate the situational awareness of pedestrians under various multitasking conditions using eye-tracking and behavior-monitoring technologies.

2.1.1 Simulation Environment

A high-fidelity pedestrian simulator was developed using a 3D virtual environment creation platform (e.g., Unity or Unreal Engine). The simulation environment replicated typical urban road-crossing settings, including:

- Two-lane and four-lane roads
- Zebra crossings with and without traffic signals
- Roadside infrastructure such as billboards, parked vehicles, and footpaths
- Dynamic traffic flow with approaching vehicles at varying speeds and densities
- Daytime lighting and ambient street noise to create a realistic atmosphere

The virtual environment allowed manipulation of visual and auditory stimuli, enabling the study of distraction effects from mobile phone use, environmental clutter, and billboards.

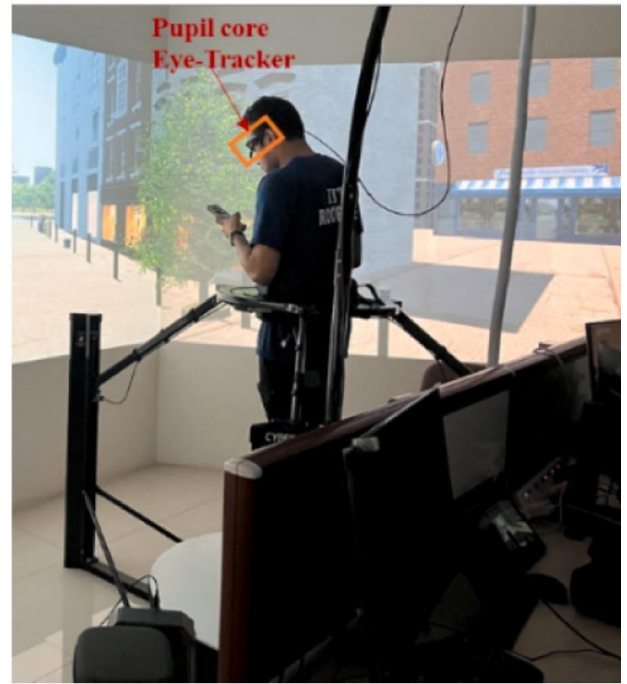
2.1.2 Participant Setup

Participants were required to walk in place or remain seated while navigating the pedestrian simulator using a gamepad or treadmill-integrated input system. Their natural gaze and head movements were preserved to simulate realistic crossing behavior.

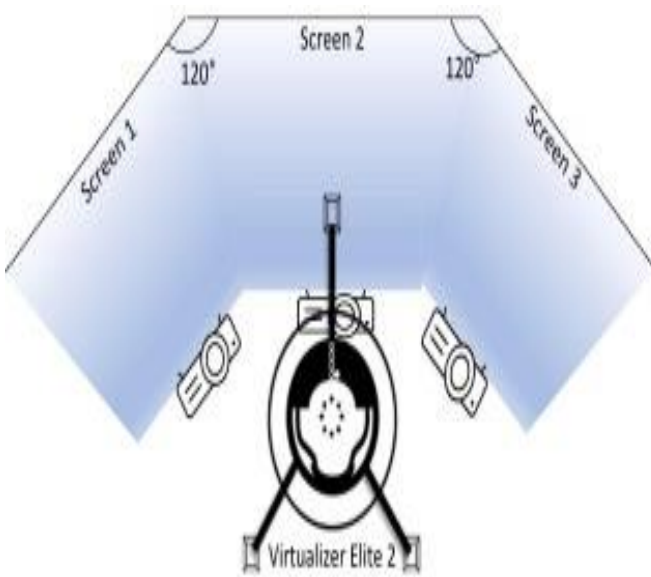
Each participant underwent a calibration procedure before the experiment to ensure accurate eye-tracking data collection.



(a)



(b)



(a)



(b)



3 Experimental Conditions

The study investigates pedestrian situational awareness under different road-crossing conditions, categorized based on the presence or absence of traffic signals.

3.1 No Countermeasure (No_CM) Condition

The No_CM condition serves as the baseline or control group in this study. Under this scenario, no additional safety countermeasures are implemented beyond standard road infrastructure. All forms of pedestrian distractions are present, including smartphone use, headphone usage, and other visual or cognitive distractions.

This setup represents typical, real-world scenarios where pedestrians often cross streets without any enhanced safety features. The primary purpose of this condition is to offer a reference point for evaluating the effectiveness of other intervention strategies. By analyzing pedestrian behavior and situational awareness in the absence of countermeasures, the study can better quantify the impact of distraction and the relative improvement brought by safety enhancements.

Key characteristics:

- All types of distractions allowed
- No visual or auditory warnings beyond standard signals
- Acts as a baseline for comparison with other safety treatments
- Helps isolate the influence of added interventions by contrast

3.2 Illuminated Crosswalks (IC) Condition

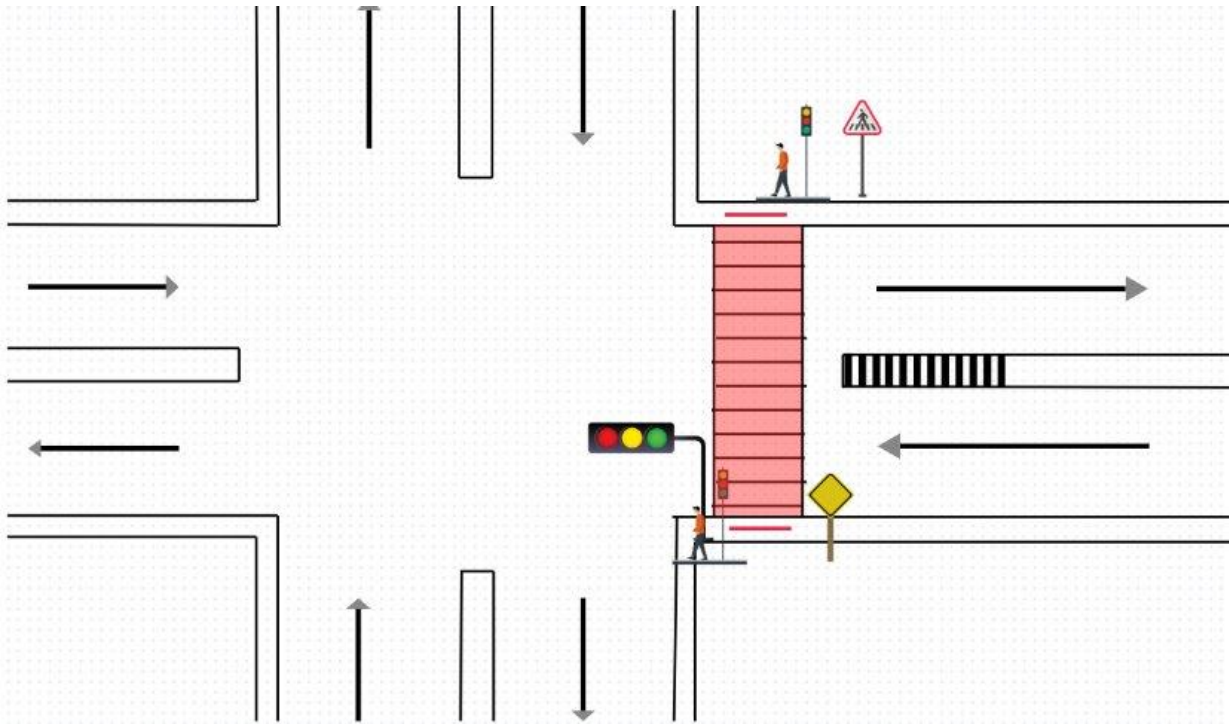
The Illuminated Crosswalks condition introduces infrastructure-based safety enhancements in the form of visual cues—such as in-ground LED lights or flashing beacons—in addition to the existing traffic signals. These crosswalks are designed to increase the visibility of the crossing zone, especially under low-light conditions or when drivers might otherwise fail to notice a pedestrian.

This condition is particularly focused on improving pedestrian detection and awareness for both walkers and approaching drivers. While the lights enhance safety by providing a clear and immediate visual alert, pedestrians are still required to exercise judgment and attention before crossing. The illuminated crosswalks aim to reduce ambiguity about right-of-way and improve crossing behavior without relying on active decision-making systems.

Key characteristics:

- Supplemental visual alerts (e.g., LED-embedded pavement lights)
- Designed to enhance pedestrian visibility and cue drivers
- Encourages safer crossing behavior through improved infrastructure
- Still depends on pedestrian decision-making and attention

These two conditions help in understanding how different environmental cues—or the lack



thereof—influence pedestrian behavior, especially when distractions are introduced. The comparison between the No_CM and Illuminated Crosswalks conditions enables a detailed assessment of how effective visual infrastructure improvements are in mitigating the risks associated with distracted pedestrian behavior.

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4 Distraction Conditions

To simulate distracted conditions, participants were exposed to one of the following multitasking activities during their crossing task:

- Undistracted baseline condition (Baseline)
- Observing a dynamic billboard display located near the crosswalk (BB)
- Listening to music via earphones (LM)
- Talking on the phone (TLK)
- Texting on a mobile phone (TXT)

Each condition was tested in randomized order to minimize learning bias.

To eliminate order effects, the sequence of trials was randomized for each participant. Importantly, a consistent traffic scenario—featuring a fixed pattern of vehicle and pedestrian movements—was used across all trials to ensure uniformity in the virtual environment. This allowed for an accurate comparison of gaze behaviour across different multitasking conditions. Each trial lasted approximately 1.5 minutes, followed by a mandatory 2-minute rest period to prevent fatigue and maintain participant engagement.



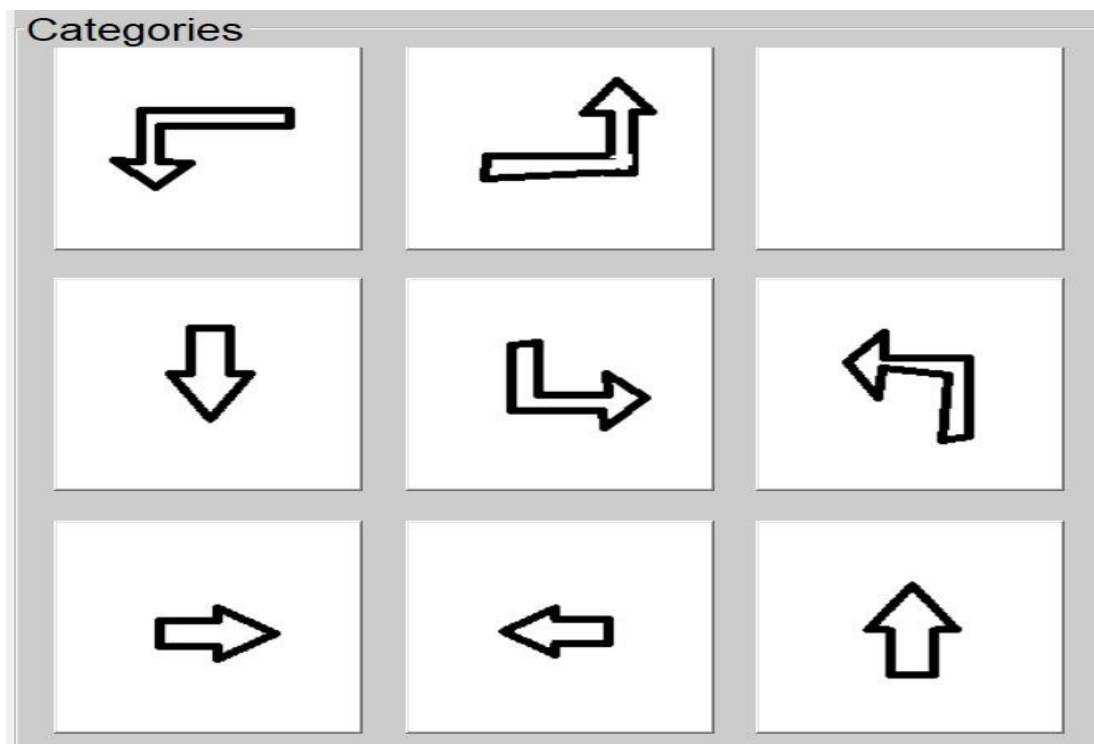
S.No.	Type of Activity	Label	Description of Activity
1.	No Multi-tasking	Baseline	Participants were not involved in any multi-tasking activity (i.e., baseline)
2.	Texting on Mobile Phone	TXT	Participants were instructed to respond to the questions asked by the experimenter via text messages on phone
3.	Talking on a mobile phone	TLK	Participants were instructed to answer the questions asked by the experimenter via voice call
4.	Listening to music	LM	Participants were listening to the music via earphones
5.	Gazing to Billboard	BB	Participants were asked to read the information displayed on the billboard.

For the texting (TXT) and talking (TLK) conditions, structured scripts were developed to emulate natural conversations between two unfamiliar individuals. These scripts included everyday topics such as education, hobbies, favourite foods, restaurants, movies, and popular tourist destinations. In the TLK condition, participants received and responded to simulated phone calls using a hand-held mobile device.

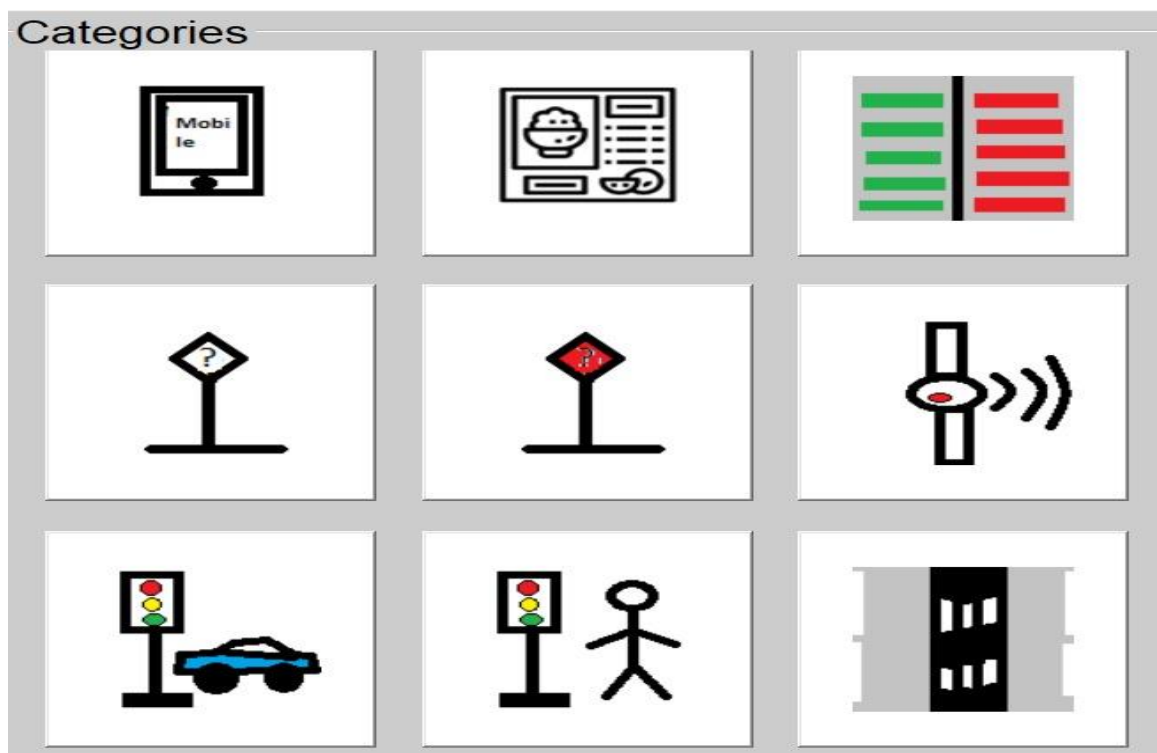
In the listening to music (LM) condition, participants were instructed to choose a song of their preference and walk while listening through earphones, allowing for a more personalized and immersive experience. For the gazing at billboard (BB) condition, a virtual advertisement billboard was positioned on the left side of the crosswalk, as shown in Figure. This billboard was exclusively present during the GBB trial and removed in all other conditions. Participants were instructed to read and process the information displayed on the billboard while crossing the road.

5 Areas of Interest (AOI)s

- Two-Wheeler (2W)
- Four-Wheeler (4W)
- Heavy Vehicle (HV)

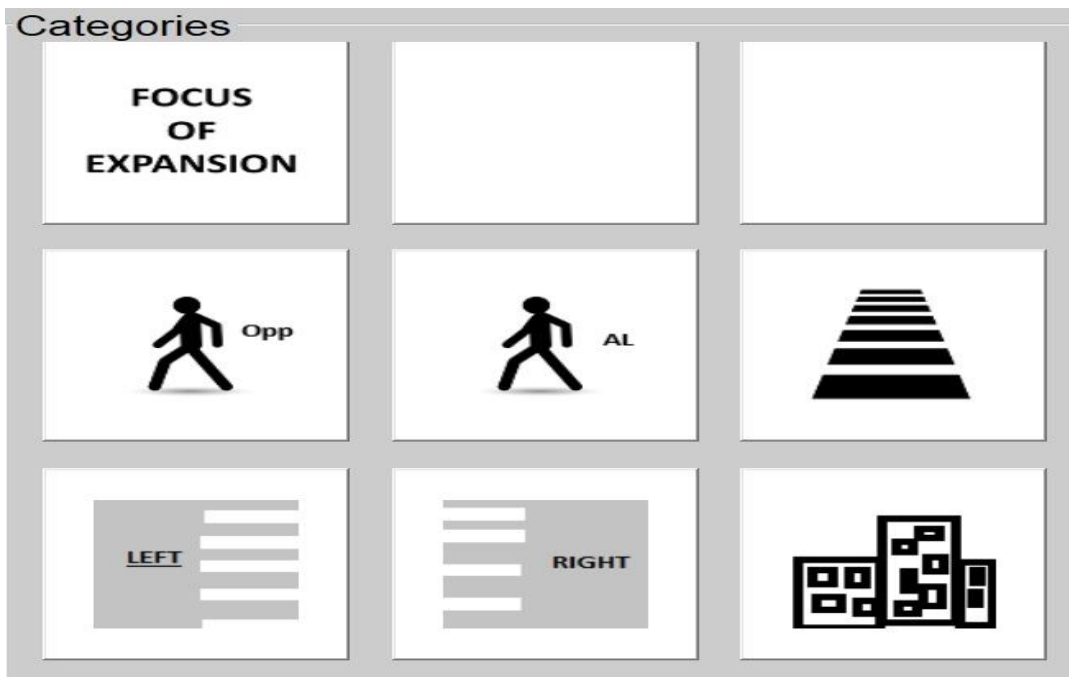


- Infrastructure Related (IR)





- Remaining



6 Data extraction

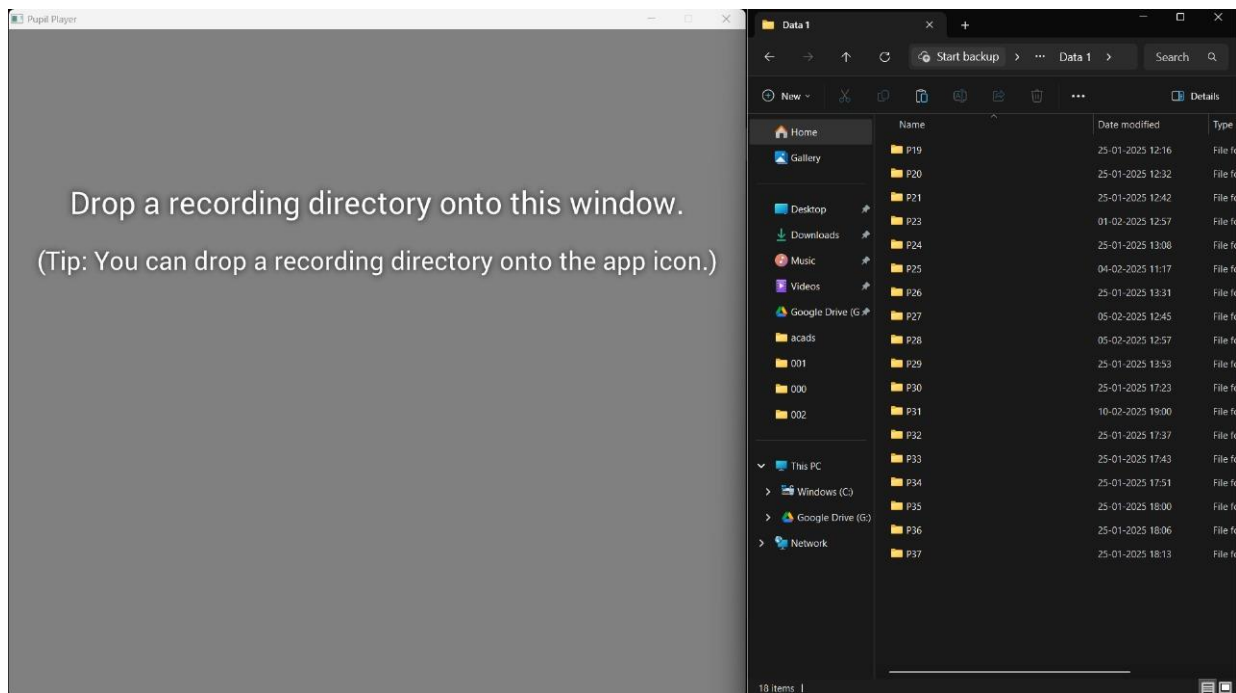
Steps Followed for Data Extraction and Labelling using gazeCode.m:

- **Initialized the MATLAB environment** by clearing previous data, closing open figures, and setting up necessary directory paths for input data, results, and category images.

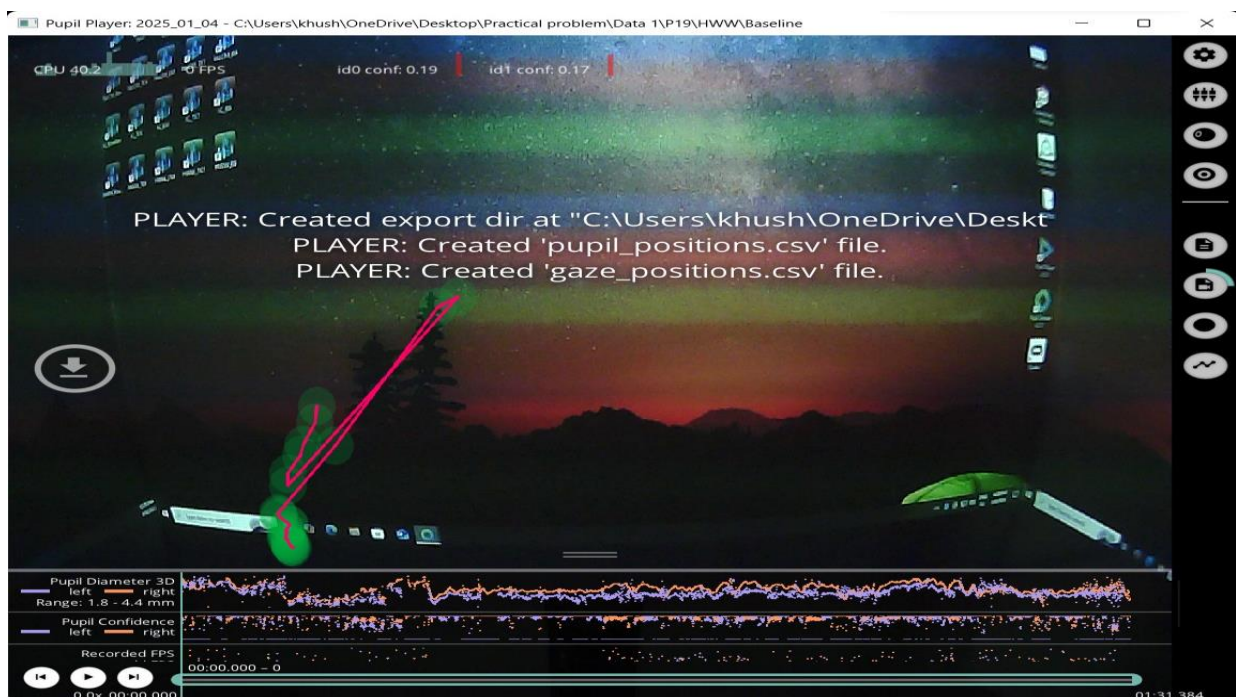
```

function gazeCode(settings)
%
% GazeCode (GlassesViewer integrated + Pupil Labs Invisible supported version)
%
% See readme.md for usage details
%
% This open-source toolbox has been developed by J.S. Benjamins, R.S.
% Hessels and I.T.C. Hooge. When you use it, please cite:
%
% Jeroen S. Benjamins, Roy S. Hessels, and Ignace T. C. Hooge. 2018.
% GazeCode: open-source software for manual mapping of mobile eye-tracking
% data. In Proceedings of the 2018 ACM Symposium on Eye Tracking Research &
% Applications (ETRA '18). ACM, New York, NY, USA, Article 54, 4 pages.
% DOI: https://doi.org/10.1145/3204493.3204568
%
% For importing data from Tobii Glasses 2 and 3, it uses GlassesViewer. When
% using this toolbox with Tobii Glasses data, please also cite
% Niehorster, D.C., Hessels, R.S., and Benjamins, J.S. (2020).
% GlassesViewer: Open-source software for viewing and analyzing data from
% the Tobii Pro Glasses 2 eye tracker. Behavior Research Methods, 52,
% 1244–1253.
% DOI: https://doi.org/10.3758/s13428-019-01314-1
%
% For more information, questions, or to check whether we have updated to a
% better version, e-mail: j.s.benjamins@uu.nl. GazeCode is available from
% www.github.com/jsbenjamins/gazeCode and GlassesViewer from
% https://github.com/dcnieho/glassesviewer
%
% Most parts of GazeCode are licensed under the Creative Commons
% Attribution 4.0 (CC BY 4.0) license. Some functions are under MIT
% license, and some may be under other licenses.
%
% Tested on:
% Matlab 2021b on Mac OSX 12.2.1
% Matlab 2022b, on Windows 11
  
```

- Selected the eye-tracker model Pupil Labs Invisible through a user prompt. This ensured that the script used the correct file structure and settings for that device.

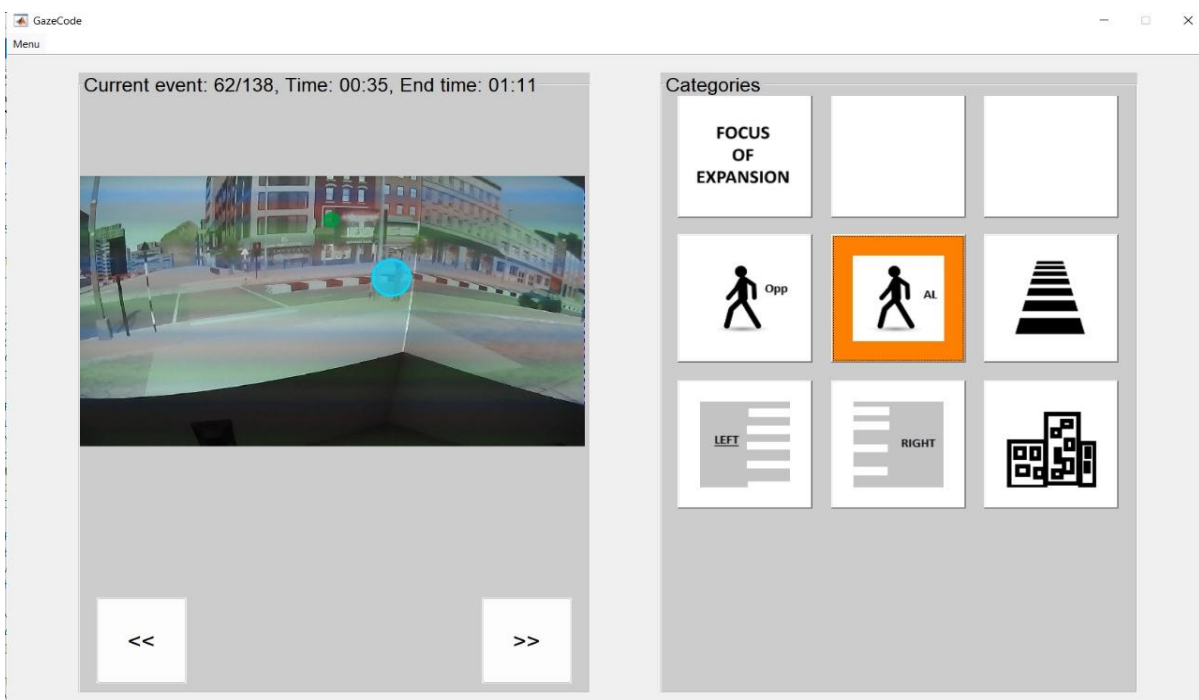


- Loaded scene video and gaze data using Video-Reader and file selection dialogs. The gaze data was typically in CSV or JSON format, containing time-stamped x and y coordinates.





- **Normalized and resampled gaze coordinates** to 200 Hz to standardize the data. The x and y positions were also converted from normalized units to actual pixel positions based on the resolution of the world camera.
- **Applied fixation detection** using either the Hooke & Camps (2013). These methods identified the start and end timestamps of each fixation based on gaze stability.
- **Calculated fixation-level metrics**, including:
 - Fixation start and end times,
 - Duration of each fixation,
 - Start and end coordinates,
 - Mean and standard deviation of x and y positions during each fixation.
- **Mapped fixations to corresponding video frames** using timestamps and computed the mid-frame of each fixation for visualization in the GUI.
- **Launched a graphical user interface (GUI)** that displayed the video frame and provided buttons for manual labelling. Each button corresponded to a category, with optional image icons for visual aid.
- **Labelled fixations manually** by clicking the category buttons or using predefined keyboard shortcuts. The label was recorded against each fixation event.



- **Stored all fixation data and labels** in a structured format, which included fixation number, timing, gaze coordinates, and assigned label.

- **Exported the final dataset** in both. mat and .xls formats, making it ready for statistical analysis, modelling, or integration with other tools like Glasses Viewer.

7 Data Transformation

After extraction, eye-tracking data was organized into a structured format by merging all five conditions for each participant across three different fields of study. This allowed for consistent analysis of gaze behavior under varied distraction scenarios.

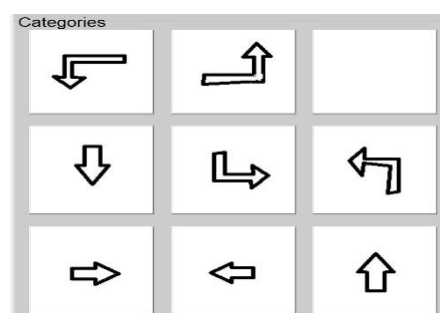
Area of Interest (AOI) Categorization

To systematically evaluate the **visual attention and situational awareness of pedestrians**, the environment was segmented into specific **Areas of Interest (AOIs)**. These AOIs represent visually distinct and functionally relevant components within the pedestrian's field of view. For improved clarity, analysis, and interpretation, the numerous objects and distractions in the urban environment were **grouped into broader categories**, each encapsulating multiple similar elements. This **clubbing of AOIs** ensures that data can be analyzed more effectively and patterns of attention can be more clearly understood.

The following categories were defined for mapping AOIs:

1. Vehicle

This category includes all types of moving or stationary motor vehicles visible to the pedestrian, such as cars, buses, motorcycles, and trucks. It is a critical AOI, as vehicles are a primary source of potential danger at crossings and intersections.



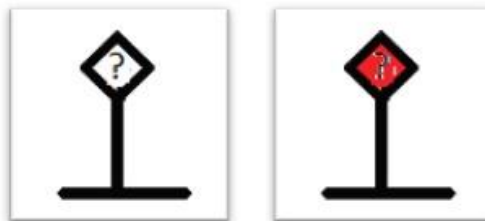
2. Traffic Signals

Includes traffic lights and pedestrian crossing signals. These elements are essential for guiding safe crossing behavior and determining the right-of-way. Pedestrian awareness of these signals is indicative of rule compliance and situational awareness.



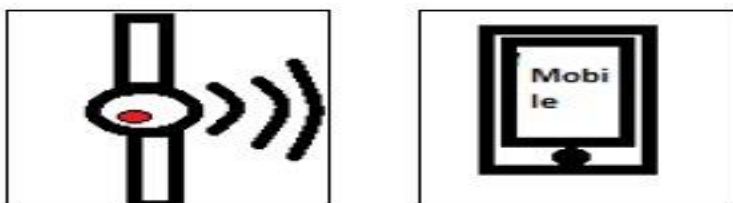
3. Sign Boards

Refers to road signs including speed limits, pedestrian warnings, lane indicators, and directional signs. These signs help inform both pedestrians and drivers about traffic conditions and crossing rules, contributing to overall road safety.



4. Gadgets

This category captures personal electronic devices used by pedestrians, such as smartphones, tablets, or headphones. As a direct source of cognitive and visual distraction, gadgets are a crucial AOI when assessing distracted behavior.



5. Billboards

Includes large advertising boards or screens placed along the road. While not inherently functional for navigation, billboards can significantly attract visual attention and distract pedestrians from monitoring their surroundings.



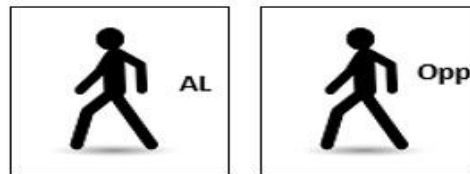
6. Buildings and Structures

Comprises nearby buildings, architectural features, and urban infrastructure not directly related to the road system (e.g., shops, offices, overpasses). This AOI helps identify how environmental context and visual clutter affect attention.



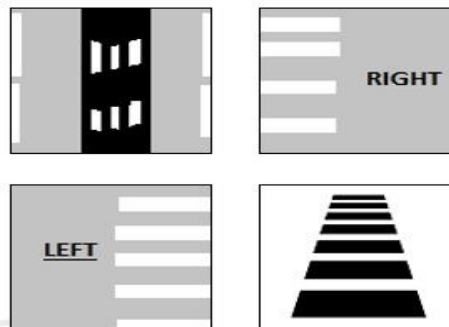
7. Fellow Pedestrians

Covers the presence and actions of other people in the vicinity, whether walking in the same direction, crossing the road, or standing nearby. Observing others can influence pedestrian decisions, particularly in ambiguous situations.



8. Road Elements

Includes elements such as road markings, curbs, sidewalks, medians, and tactile paving. These structural cues play a key role in guiding pedestrian movement and help assess attention to path-related details.



By organizing the AOIs into these eight broader categories, the study enhances its ability to:

- **Identify patterns in gaze behavior**
- **Quantify distraction sources**
- **Analyze attention allocation**
- **Evaluate the influence of environmental design on pedestrian safety**

This structured approach enables a **more accurate and insightful interpretation** of eye-tracking or observational data, ultimately contributing to more informed recommendations for pedestrian safety interventions.

8.Results

8.1 Fix_Count by Type of AOI (IC vs NO_CM)

- **Image 1: "Fixation Count by Type of AOI – NO_CM"**

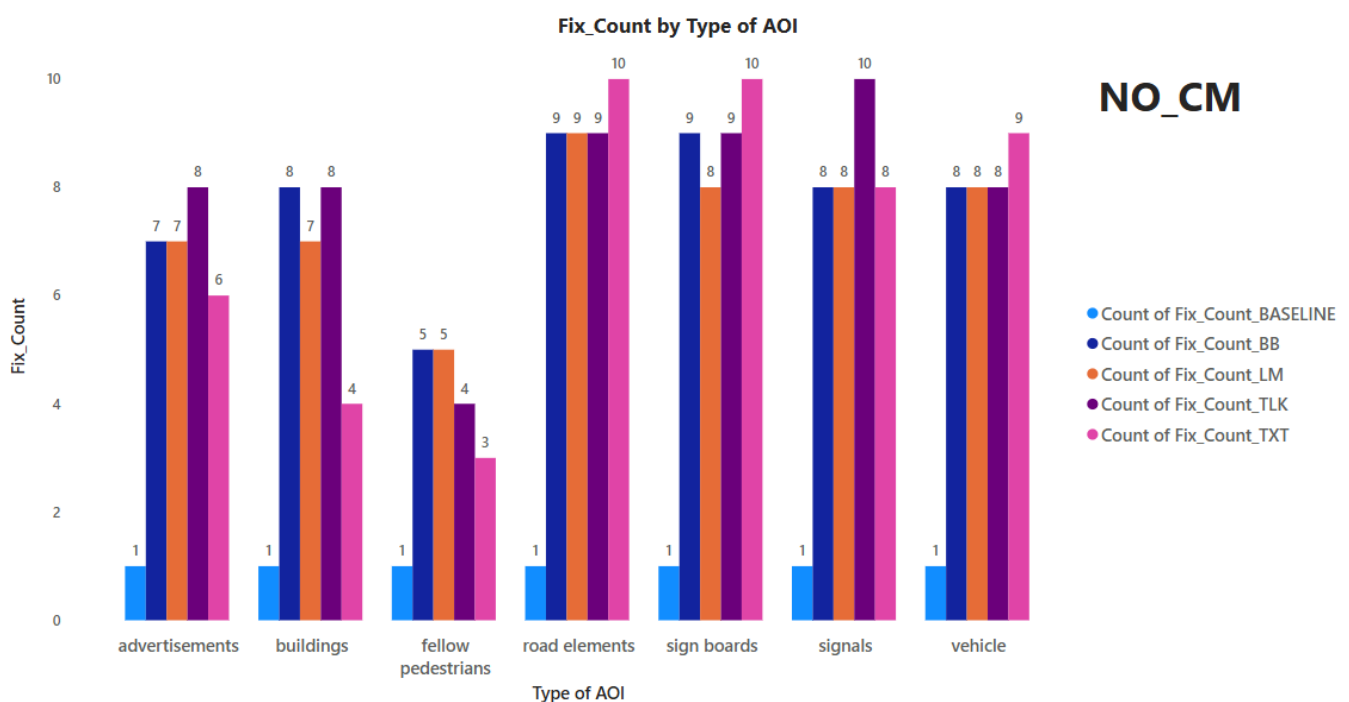
This bar chart compares fixation count across different Areas of Interest (AOIs) under the No Countermeasure (NO_CM) condition. The AOIs include advertisements, buildings, fellow pedestrians, road elements, sign boards, signals, and vehicles.

Distractions increase fixation count: Across most AOIs, distracted conditions (especially TLK and TXT) show higher fixation counts compared to the baseline. This suggests increased cognitive load and scanning behavior under distraction.

Road elements and sign boards had the highest fixation counts (up to 10) in all distracted conditions — indicating critical attention zones during crossing.

Advertisements and buildings had relatively fewer fixations — but higher during distraction (especially BB and LM), which shows non-task-related attention diversion.

Fellow pedestrians had a notable drop in fixation during TXT (count = 3) and TLK (count = 4), suggesting reduced social/spatial awareness under these distractions.



• **Image 2: "Fixation Count by Type of AOI – IC (Illuminated Crosswalk)"**

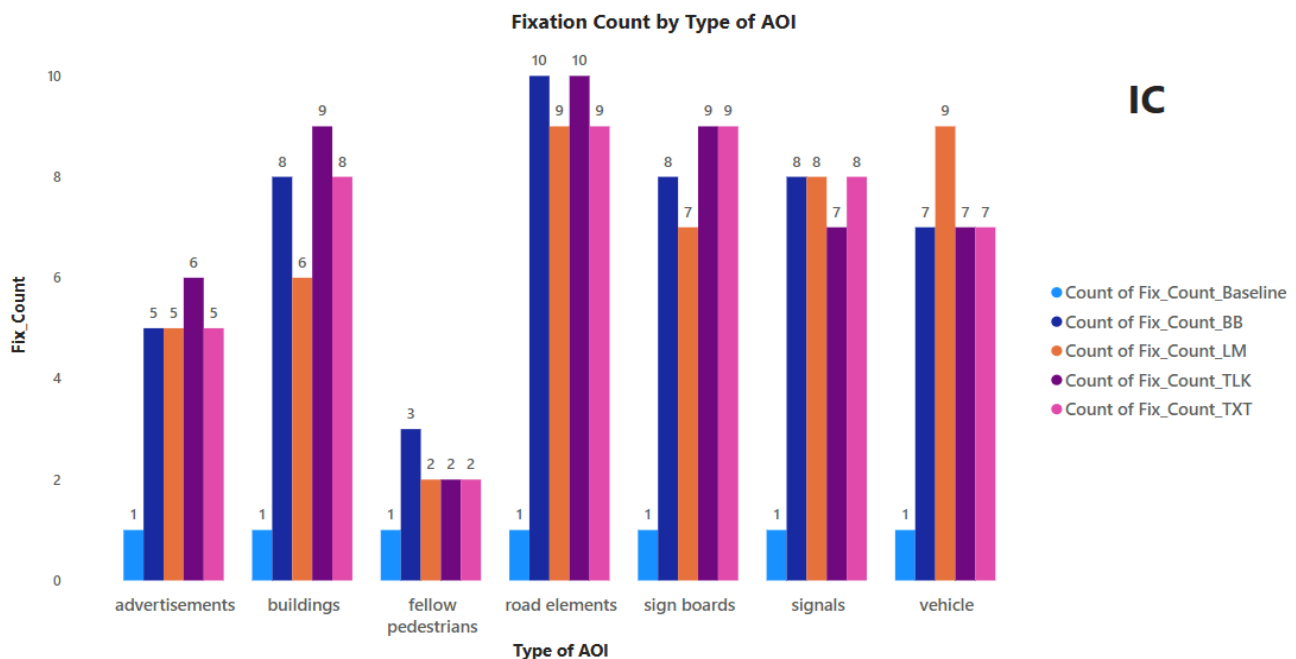
This chart follows the same structure as the first but under the Illuminated Crosswalk (IC) condition, which introduces visual cues to improve safety.

More uniform fixation counts: Across AOIs, fixation counts are more consistent and slightly lower compared to NO_CM, showing that visual aids reduce scanning effort and cognitive burden.

Fixation on road elements and buildings remains high, indicating these AOIs are still primary focus zones even with safety measures.

Distractions still increase fixation compared to baseline, but the difference is less pronounced, suggesting that IC mitigates distraction effects.

Fellow pedestrians received low fixation (count = 2 or 3) across distractions — reinforcing that social awareness remains compromised even under IC when distracted.



Overall Insight:

- Distraction increases fixation counts, indicating more scanning and cognitive load.
- Illuminated Crosswalks (IC) help stabilize gaze behavior by making critical cues more salient.
- Despite IC, mobile-based distractions (TXT, TLK) still reduce awareness of dynamic elements like fellow pedestrians.
- Designing crosswalks with strong visual cues can partially compensate for distracted behavior but cannot eliminate all risks.

8.2 Fix_Dur(ms)_Baseline by Type of AOI (IC vs NO_CM)

Illuminated Crosswalk (IC)

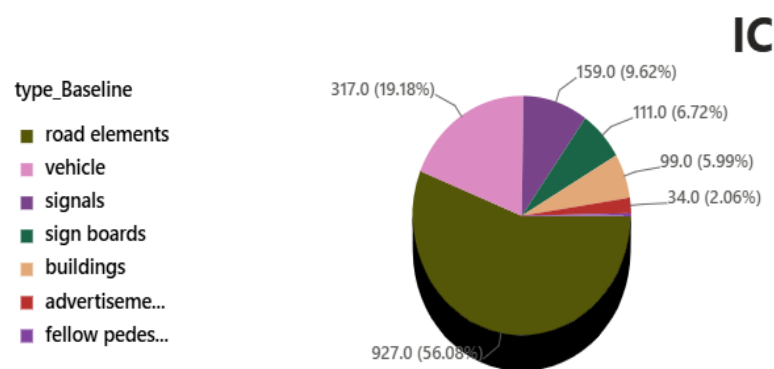
Road elements received the highest attention: 56% of total fixation duration (927 ms), suggesting that visual cues draw focus to the most relevant crossing areas.

Buildings (19.18%) and vehicles (6.72%) followed, indicating moderate peripheral scanning.

Signals and sign boards had minimal fixation durations (~6% and ~2%), which may reflect redundancy due to clear illuminated paths.

Overall pattern: Gaze is efficiently concentrated on navigation-relevant zones, reducing distractions.

Count of fix_dur (ms)_Baseline by type_Baseline



No Countermeasure (NO_CM)

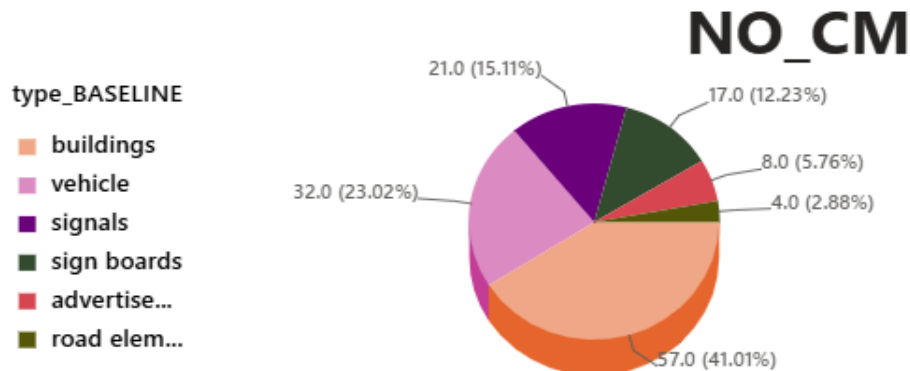
Road elements had lower relative attention: only 41% of fixation duration (57 ms).

More spread-out attention: advertisements (23%), buildings (15%), and signals (12%) received significant gaze time.

Indicates dispersed visual attention, likely due to lack of guiding cues, making pedestrians rely on more environmental scanning.



Count of fix_dur (ms)_BASELINE by type_BASELINE



Comparison Insight

- IC significantly improves visual focus on road elements, reducing attention to irrelevant or less important areas like advertisements.
- NO_CM condition shows higher distraction toward non-critical AOIs, which may increase risk during crossing.
- This highlights the effectiveness of IC in improving situational awareness and gaze prioritization during pedestrian movement.

8.3 Fix_Dur(ms)_BB by Type of AOI (IC vs NO_CM)

Illuminated Crosswalk (IC)

Road elements still dominate attention with 28.59% fixation time (629 ms), showing that even under billboard distraction, IC helps pedestrians stay focused on critical elements.

Advertisements (26.18%) and buildings (16.91%) also attract significant gaze, as expected under BB distraction.

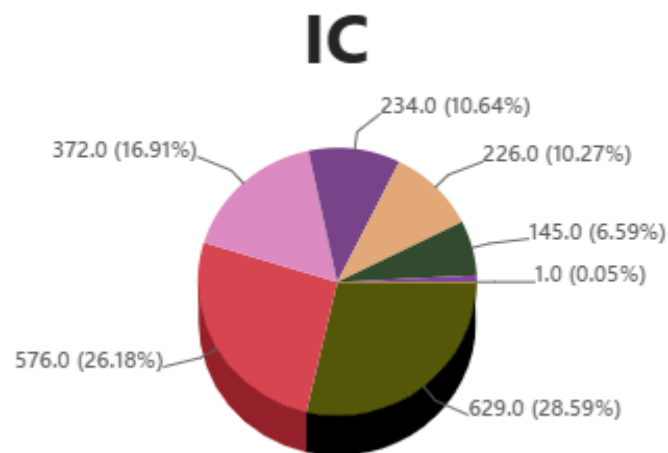
Signals and vehicles receive moderate attention (~10% each), while tech (mobile use) and fellow pedestrians are almost negligible.

Key takeaway: IC retains substantial attention toward road-critical cues, reducing over-reliance on environmental scanning despite distraction.

Count of fix_dur (ms)_BB by type_BB

type_BB

- road elem...
- advertise...
- vehicle
- signals
- buildings
- sign boards
- fellow pe...
- tech

**No Countermeasure (NO_CM)**

Advertisements now receive the most attention: 28.11% of fixation time (432 ms), showing that pedestrians are highly distracted under BB without guidance.

Buildings (18.28%) and signals (17.37%) also take up attention, indicating diversion to surrounding elements.

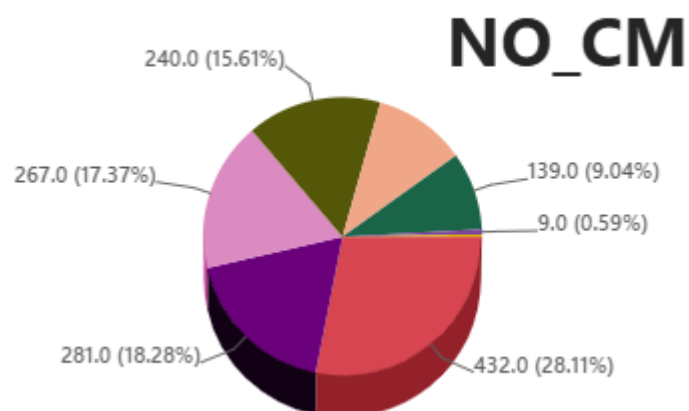
Road elements get only 9% of fixation duration (139 ms), reflecting poor situational awareness.

Overall pattern: Without IC, the billboard distraction overwhelms attention, pulling gaze away from critical crossing zones.

Count of fix_dur (ms)_BB by type_BB

type_BB

- advertiseme...
- signals
- vehicle
- road elemen...
- buildings
- sign boards
- fellow pede...
- tech



Comparison Insight

- Under BB distraction, IC preserves focus on road elements, while NO_CM leads to excessive attention on ads and surroundings.
- This underscores that IC can partially counteract visual distractions, keeping pedestrians more situationally aware.

8.4 Fix_Dur(ms)_LM by Type of AOI (IC vs NO_CM)

Illuminated Crosswalk (IC)

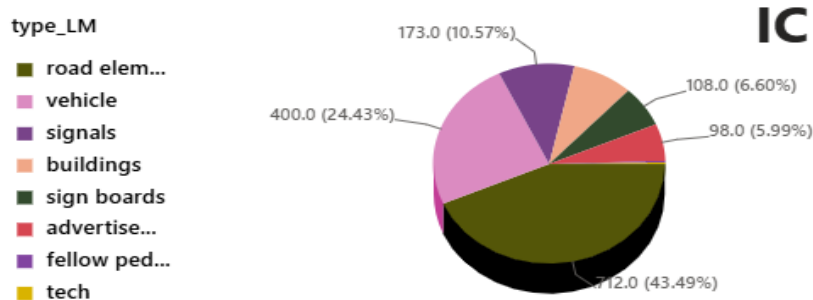
Road elements again dominate fixation duration with 43.49% (720 ms) — confirming IC's strong role in directing pedestrian attention to key areas even during auditory distraction.

Buildings (24.43%) and vehicles (10.57%) also receive substantial attention, showing a moderate degree of environmental scanning.

Signals and sign boards get relatively low attention (~6% each).

Overall observation: Even though the auditory distraction (LM) doesn't directly compete visually, IC effectively maintains focus on road-relevant elements.

Count of fix_dur (ms)_LM by type_LM



No Countermeasure (NO_CM)

Buildings (31.09%) and advertisements (17.92%) attract more attention than road elements.

Road elements receive only 17.05% of fixation duration — significantly lower than IC.

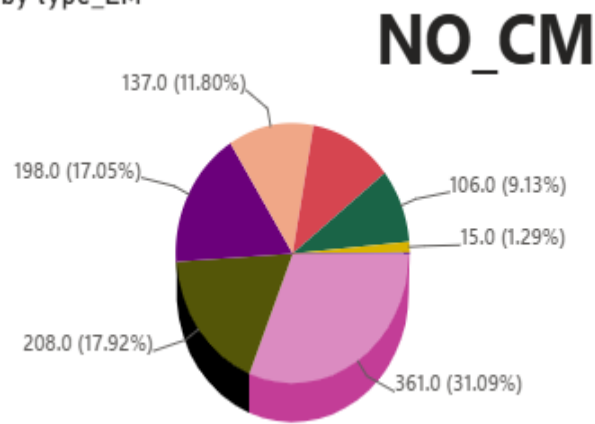
Tech and fellow pedestrians also pull gaze (9.13% and 11.80%), indicating dispersed attention.

Conclusion: Without IC, the listening distraction leads to more scattered gaze and less awareness of critical cues.

Count of fix_dur (ms)_LM by type_LM

type_LM

- vehicle
- road eleme...
- signals
- buildings
- advertisem...
- sign boards
- tech
- fellow ped...



Comparison Insight

- Under LM distraction, IC maintains focused attention on road elements (43.5%), while NO_CM shows a shift to buildings and ads.
- This confirms that IC helps limit passive drift in attention during non-visual distractions, improving overall pedestrian safety.

8.5 Fix_Dur(ms)_TLK by Type of AOI (IC vs NO_CM)

Illuminated Crosswalk (IC)

Road elements remain the dominant focus, with 37.13% (636 ms) of fixation time — indicating IC's consistent support for attention guidance even under high cognitive load.

Vehicles (23.41%) and buildings (12.20%) also attract substantial gaze time, reflecting some attention to the broader environment.

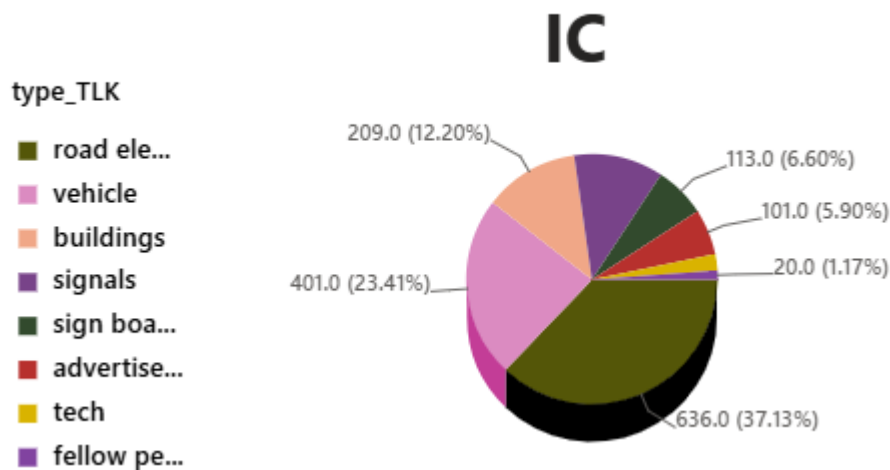
Signals (6.60%) and fellow pedestrians (5.90%) receive moderate attention.

Technology-related fixations are minimal (1.17%), as expected in a voice-only distraction.

Conclusion: Even under cognitively demanding tasks like phone calls, IC helps retain strong situational awareness.



Count of fix_dur (ms)_TLK by type_TLK



No Countermeasure (NO_CM)

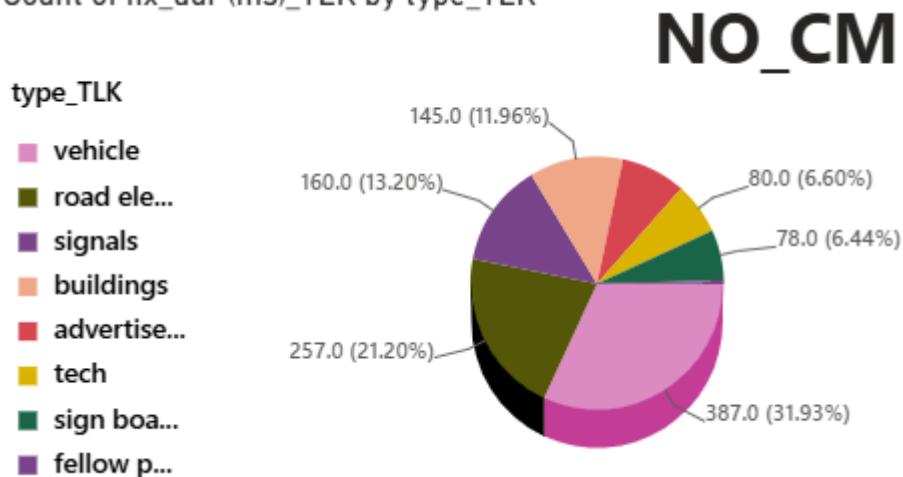
Fellow pedestrians draw the most gaze at 31.93% (387 ms), possibly due to reliance on peripheral social cues in absence of infrastructure guidance.

Tech (21.20%) and vehicles (13.20%) also gain attention, indicating cognitive wandering.

Road elements get only 11.96%, showing a clear drop in safety-critical attention.

The data shows greater visual scatter without IC when engaged in phone conversations.

Count of fix_dur (ms)_TLK by type_TLK



Comparison Insight

- IC helps maintain critical gaze focus (37%) on road elements, while NO_CM results in attention drift to irrelevant areas like fellow pedestrians and tech.
- Talking on the phone is especially disruptive, and the presence of IC significantly mitigates distraction impact.

8.6 Fix_Dur(ms)_TXT by Type of AOI (IC vs NO_CM)

IC (Illuminated Crosswalk)

Signals draw the highest gaze share at 31.71% (339 ms), suggesting strong attentional prioritization of traffic control and safety cues.

Buildings (23.85% / 255 ms) and vehicles (23.48% / 251 ms) also receive significant visual attention, indicating broad situational awareness.

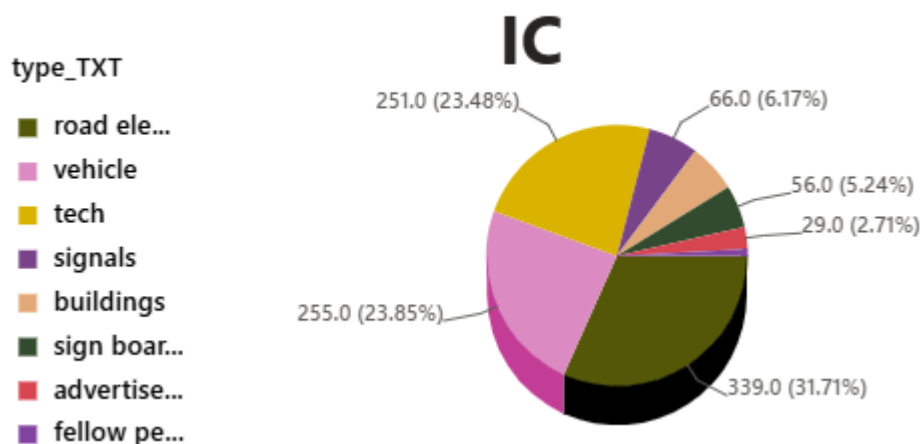
Road elements (6.17%) and technology (5.24%) receive moderate gaze time, possibly due to relevance for spatial planning and control.

Very low fixation on fellow pedestrians (2.71%), ads, or signboards — indicating minimal distraction or social scanning.

Conclusion:

Under IC, attention is effectively directed toward functional and safety-relevant elements (signals, infrastructure, traffic), reflecting an efficient, environment-attuned gaze strategy.

Count of fix_dur (ms)_TXT by type_TXT



NO_CM (No Countermeasure)

Fellow pedestrians dominate fixation with 45.94% (490 ms) — suggesting high social cue dependency in the absence of visual guidance.

Buildings (18.51%) and technology (14.36%) also attract strong attention, potentially reflecting cognitive drift or environmental curiosity.

Road elements (6.04%) and signals (5.74%) receive minimal attention — a notable drop in safety-critical focus.

Ads and signboards continue to draw almost no attention.

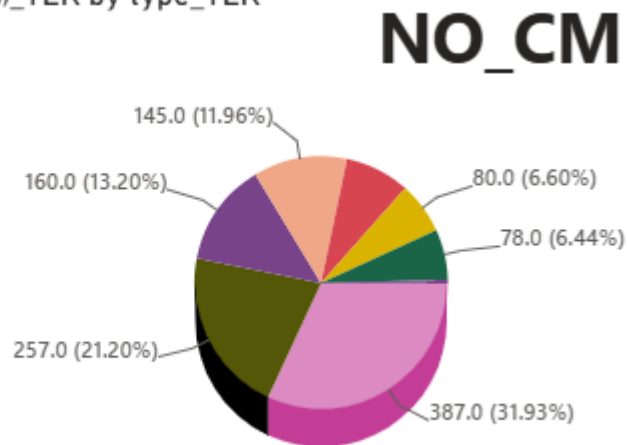
Conclusion:

In the absence of IC, gaze behavior shifts toward social and non-functional elements, indicating reduced hazard anticipation and increased cognitive distraction.

Count of fix_dur (ms)_TLK by type_TLK

type_TLK

- vehicle
- road ele...
- signals
- buildings
- advertise...
- tech
- sign boa...
- fellow p...



Comparison Insight

- IC preserves critical focus on signals (32%) and vehicles/buildings (~24%), supporting navigation and safety.
- NO_CM conditions lead to a shift toward social stimuli — fellow pedestrians (46%) and tech (14%), weakening traffic awareness.
- This indicates that IC effectively anchors gaze to relevant safety elements, while NO_CM allows gaze fragmentation.

8.7 Fix_Count by Type of AOI Considering Education (IC vs NO_CM)

IC (Illuminated Crosswalk)

Road elements dominate fixation count at 931 total fixations (509 PG + 222 Schooling + 200 UG) – showing strong environment-focused attention across education levels.

Vehicles follow with 355 fixations, indicating attention toward dynamic road users.

Signals (126), signboards (89), and buildings (99) receive moderate attention, supporting situational awareness.

Advertisements (22) and fellow pedestrians (negligible) receive minimal fixations, suggesting low distraction or social focus.

Education-wise Patterns:

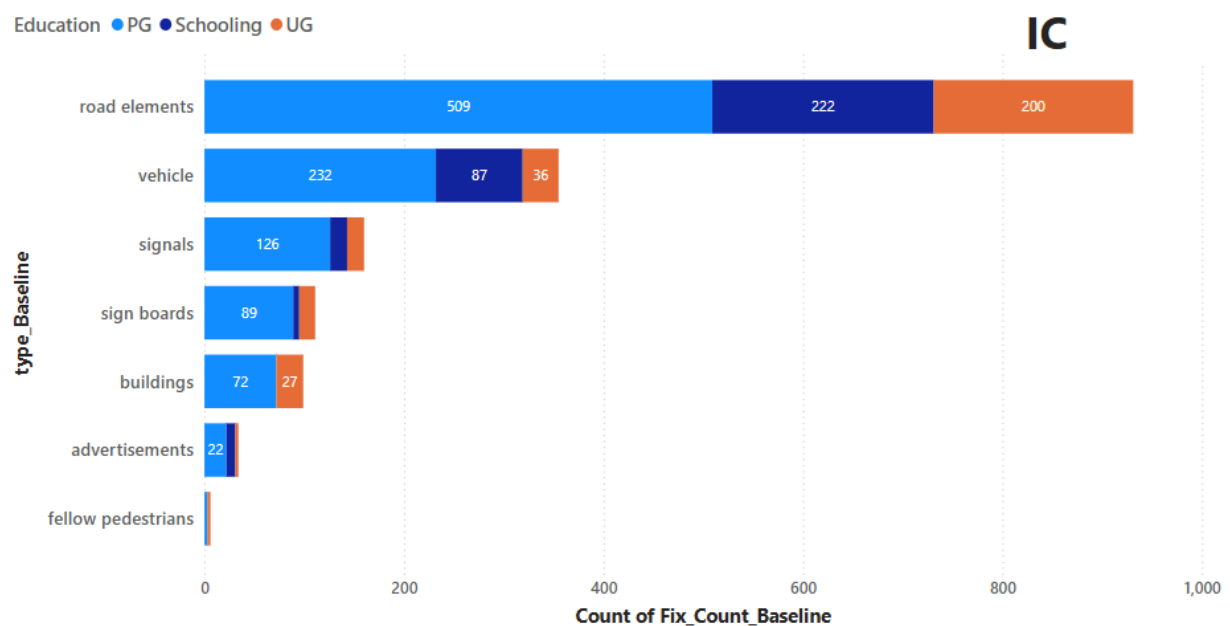
Postgraduates (PG) consistently show highest fixation counts across all categories, implying higher attentional engagement and scanning under IC.

Schooling and UG show decreasing gaze counts, but still prioritize road elements and vehicles.

Conclusion:

With IC, fixation behavior is safety-aligned, especially for PG participants who show heightened awareness of road elements and vehicles. Distraction elements like ads or social presence are largely ignored.

Count of Fix_Count_Baseline by type_Baseline and Education





NO_CM (No Countermeasure)

Vehicles attract the most attention with 419 fixations (336 PG + 50 Schooling + 33 UG), suggesting motion-based attention without structured guidance.

Road elements drop to 311 total fixations, reflecting weaker orientation to static infrastructure.

Buildings (181) and signals (154) gain moderate attention, potentially reflecting passive environmental scanning.

Higher distractions: Ads (69), fellow pedestrians (19), and tech (12) — all receiving more attention than under IC.

Education-wise Patterns:

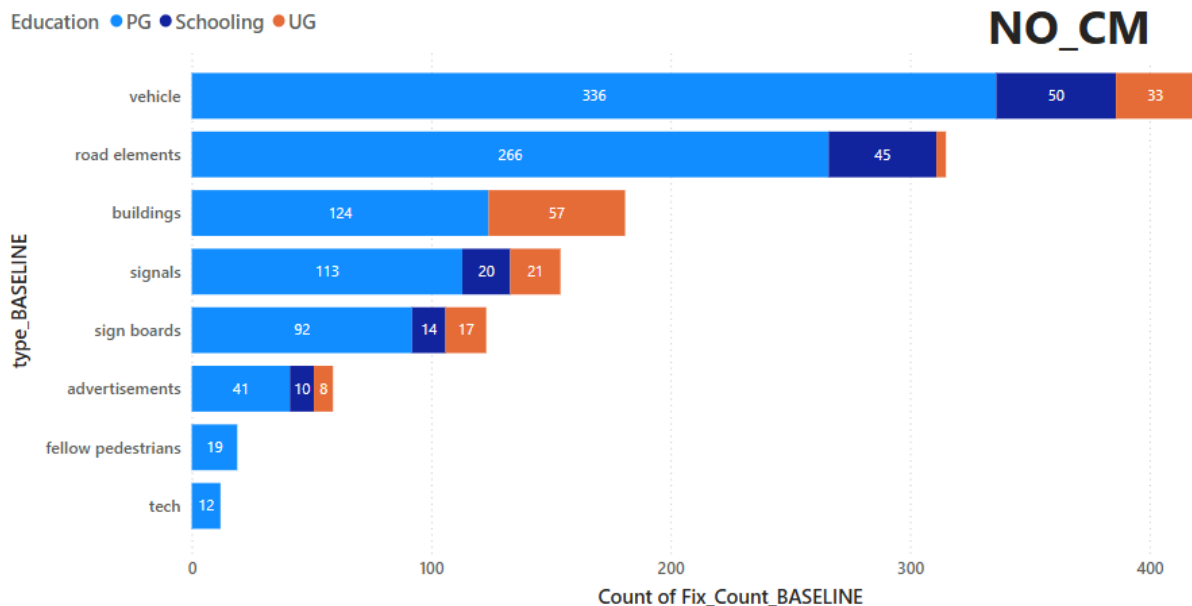
PG participants again dominate fixation counts, especially toward vehicles and infrastructure.

Schooling and UG groups show increasing gaze to non-critical elements (ads, buildings), hinting at cognitive drift or distraction in absence of IC.

Conclusion:

Without IC, fixation behavior becomes more reactive and scattered, focusing more on vehicles and buildings, and less on safety cues like road elements and signals. There's a measurable rise in attention to distractions, particularly in less-educated groups.

Count of Fix_Count_BASELINE by type_BASELINE and Education



Comparison:

Element	IC(Fix_Count)	NO_CM(Fix_Count)	Insight
Road Elements	931	311	IC boosts infrastructure awareness by 3 times
Vehicles	355	419	NO_CM promotes motion driven fixation
Signals	126	154	Slightly more fixation in NO_CM, but less structured
Advertisements	22	69	Distraction increases in NO_CM
Fellow Pedestrians	-	19	Social scanning grows without IC
Tech	0	12	Technology related fixation appears only under NO_CM

8.8 Avg_Fix_Dur(ms) vs Type of AOI (IC vs NO_CM)

The comparison between the IC and NO_CM conditions reveals substantial differences in average fixation duration (AFD) across various Areas of Interest (AOIs), reflecting the influence of visual infrastructure on pedestrian attention patterns under distraction.

- **Focus on Road Elements:**

Under IC, pedestrians maintained high average fixation durations on road elements across all conditions (ranging from 387 ms to 432 ms), indicating that the illuminated infrastructure effectively guided attention toward safety-critical zones.

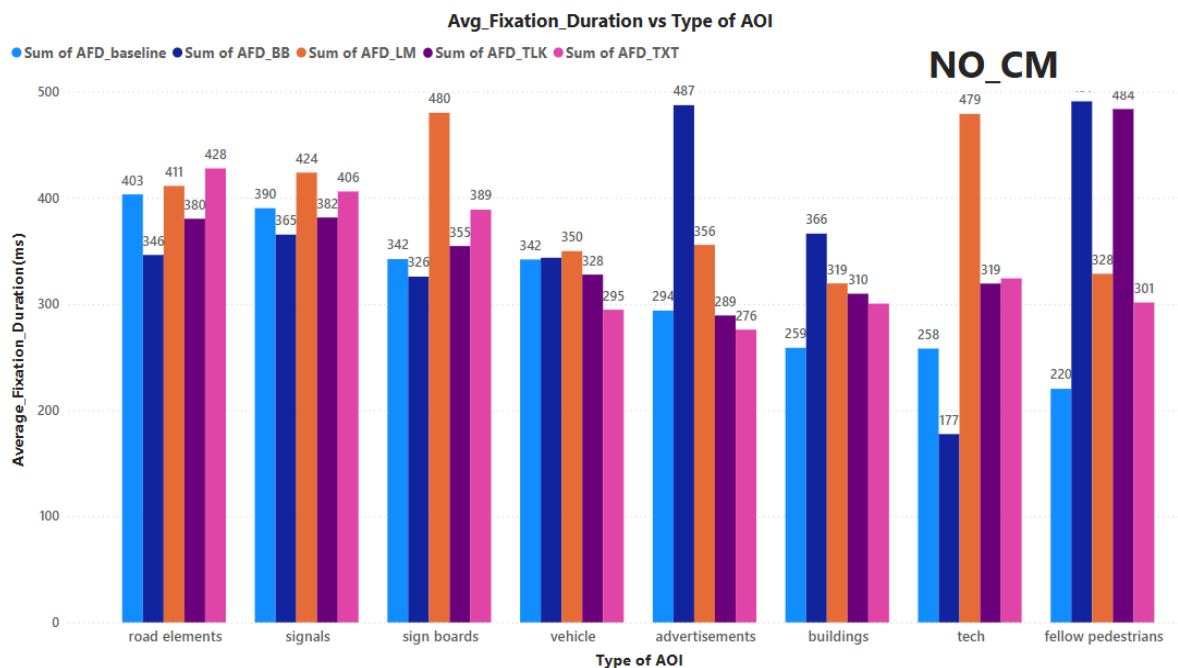
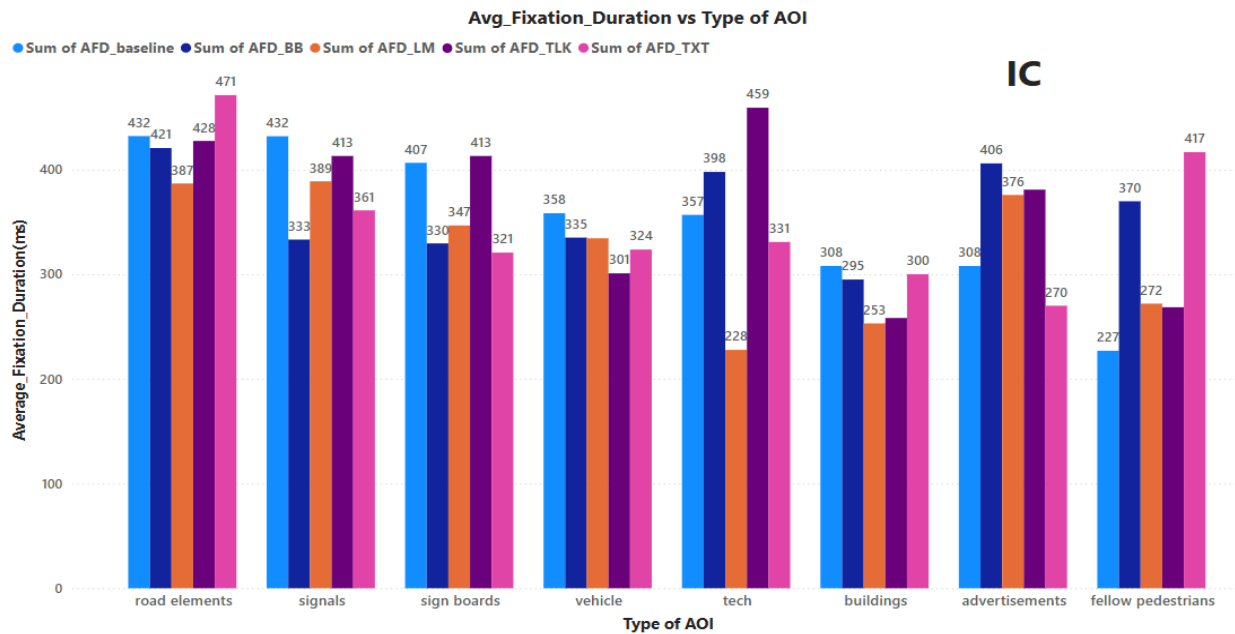
In contrast, under NO_CM, fixation on road elements dropped significantly, with values ranging from 346 ms to 411 ms, showing that in the absence of visual cues, attention was more fragmented and less safety-focused.

- **Signals and Visual Guidance:**



The signals AOI under IC showed the highest fixation in the TXT condition (471 ms), suggesting that illuminated guidance paired with distraction helps refocus attention on formal crossing cues.

Under NO_CM, while fixation on signals was also high (428 ms under TXT), the distribution was more variable and less structured, suggesting occasional reactive rather than proactive attention.



- **Distraction by Advertisements and Technology:**

In the NO_CM condition, advertisements and tech drew significantly higher fixation durations, especially in BB and LM scenarios (advertisements: 487 ms, tech: 479 ms), indicating these elements strongly diverted attention in the absence of IC.

Under IC, while these AOIs still attracted attention, the fixation durations were notably lower (advertisements max at 406 ms, tech max at 459 ms), showing that IC helped suppress excessive distraction.

- **Impact of Cognitive Distractions (TLK and TXT):**

The TLK and TXT conditions showed elevated fixation durations on fellow pedestrians and tech under NO_CM (fellow pedestrians: 484 ms, tech: 319 ms), reflecting fragmented attention and social cue reliance.

Under IC, although TLK and TXT still increased gaze on non-critical AOIs, the impact was less severe (fellow pedestrians: 417 ms in TXT, tech: 459 ms in TLK), suggesting a partial mitigation of cognitive distraction through infrastructure.

- **Environmental Scanning and Peripheral Attention:**

In the absence of IC, AOIs like buildings, sign boards, and fellow pedestrians saw increased fixation durations (e.g., sign boards: 480 ms under LM, buildings: 366 ms baseline), indicating widespread visual dispersion.

Conversely, with IC, fixation on these peripheral AOIs remained more consistent and lower in duration, indicating better visual discipline and reduced environmental scanning.

The analysis clearly demonstrates that Illuminated Crosswalks (IC) play a vital role in concentrating pedestrian attention on essential safety elements, even under various distraction scenarios. By contrast, in No Countermeasure (NO_CM) settings, pedestrian gaze becomes more scattered, with increased susceptibility to visual and cognitive distractions such as billboards, mobile devices, and surrounding pedestrians. These findings support the implementation of IC systems to enhance situational awareness and reduce accident risk in real-world pedestrian crossings.



9 Conclusions

The study confirms that Illuminated Crosswalks are an effective intervention to maintain pedestrian attention on road-critical elements even during distraction. They help anchor visual attention, reduce cognitive load, and dampen the impact of external distractors like mobile use or advertisements.

Conversely, the absence of such countermeasures in the NO_CM condition results in fragmented, socially driven, and distraction-prone gaze behavior, which can significantly elevate the risk of pedestrian accidents.

Thus, integrating infrastructure-based visual cues (such as IC) into urban traffic systems is a proven, data-backed strategy to enhance pedestrian safety, particularly in distraction-heavy modern environments.

10 References

- <https://www.sciencedirect.com/science/article/pii/S0001457524004573#f0015>
- [Endsley, M. R. \(1995\). *Toward a theory of situation awareness in dynamic systems*. Human Factors, 37\(1\), 32–64.](#)
- [Oxley, J., Ihsen, E., Fildes, B., Charlton, J., & Day, R. \(2005\). *Crossing roads safely: An experimental study of age differences in gap selection by pedestrians*. Accident Analysis & Prevention, 37\(5\), 962–971.](#)